

Yuheng Wu | Email: wyh2ah@gmail.com | Mobile: (646) 279-0528 | Brooklyn, NY 11201 | F1 Student-OPT 3Years
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Education

New York University, NY, USA

Sep.2024-May.2026

Master of Science in Computer Engineering | GPA: 4.00/4.00

Relevant Coursework: Computer Architecture (C++, C, RISC-V), Introduction to Java (SpringBoot, JavaFX), Database (SQL, MySQL, Java), Machine Learning (Python), Internet Architecture & Protocols (Python), Algorithm and Data Structure, Big Data, Deep Learning, Probability and Stochastic Processes

ShanghaiTech University, Shanghai, China

Sep.2020-Jul. 2024

Bachelor of Engineering in Computer Science | Major GPA: 3.76/4.00

Relevant Coursework: Algorithm and Data Structure (C++), Computer Architecture (C, RISC-V), Database (SQL), Operating Systems, Software Engineering (Matlab App Designer), Probability and Statistics, Digital Circuits, Human-Computer Interaction, Parallel Computing (C++, CUDA), Cryptography (C++, Python)

Honors: ShanghaiTech University 2023 - Outstanding Teaching Assistant (On Operating Systems Course)

Professional Experience

Systems Development Engineer Intern | Amazon Web Services (AWS), Annapurna Labs

May 2025-Aug.2025

AWS Trainium workload analysis and automation for silicon coverage profiling (Python, EC2, S3)

- Examined workload stress on SoC engines and address space/ISA coverage on EC2 Trainium1/2 Burn-in test payloads to model industry machine learning workload stress on hardware.
- Utilized Python and InfluxDB to automate the process to build coverage metrics from AWS neuron-tools.
- Improved the existing payloads to provide the missing coverage and designed a test & debug strategy to fix 30+ unsellable Trainium2 machines with numerical errors.

Software Development Intern | Shanghai Aerospace Yigen Intelligent Technology Co., Ltd

July.2022-Aug.2022

Performance Testing using Python & Selenium (Affiliated with China Aerospace Science and Technology Corporation)

- Responsible for function and performance testing of a production operation management platform (using Python).
- Conducted automated testing using Selenium scripts to verify system functionality and stability as well as performed load testing using Jmeter to assess system response times and throughput under high load conditions.
- Configured and deployed test hardware system and execution environment and conducted product test.

Projects & Academic Research

Researcher | Simulation and Visualization of Heart Models Based on GPU

Jan.2024-May.2024

Using Parallel Computing in CUDA C++ and OpenGL to Simulate Heart Models in Real Time

- Utilized C++11 to implement a VTK (The Visualization Toolkit) and Json format converter so that real-world patients' heart data can be smoothly run on our heart simulation model and displayed through OpenGL.
- Implemented some APIs based on ZeroMQ to emulate the stimulation of pacemakers on the heart as well as acquire the 12-lead ECG (Electrocardiogram) signal of the simulated heart to assist doctors in diagnosing illnesses.
- By using CUDA, the speed of heart simulations can be greatly accelerated, reducing the time required for a cycle of electrical signal propagation from 20 minutes to just 3 seconds, making the simulations real-time and practically valuable.

Researcher | CloudFlow: Management System for Multi-Node Cloud Computing

Summer 2023

Cloud Computing System Development using Kubernetes & Go

- Conducted an App + MySQL (multi-app deployment) experiment using Kubernetes.
- Designed and implemented a scheduler in Go for a Cloud Computing platform (CloudFlow); function of the scheduler included process management; manage applications and computing tasks; channel workers and tasks; clean up.
- The scheduler enables CloudFlow nodes to be optimally scheduled to several physical nodes, ensuring between-nodes communication so that MPI or PyTorch applications can be deployed directly to execution context.

HPC Programmer | ASC (Asia Supercomputer Community) Student Supercomputer Challenge

Summer 2023

Awarded First Prize of 10th ASC Student Supercomputer Challenge (Linux, CUDA, High Performance Computing)

- Deployed a multi-node Linux environment with specific adjustment to node topologies and power distribution across CPU cores and GPUs to achieve performance optimization for applications under 3000 Watts power limit.
- Ranked 2nd place in DeepMD application optimization sub-segment of the competition; used MPI tuning to match server topologies, profiled computing bottleneck & conducted multi-node (especially multi-GPUs) optimization, significantly reducing the DeepMD application runtime with a 3.5x speed improvement.

Skills and Interests

Programming: C++, C, CUDA C++, Java, SQL, Matlab, Go, Python, RISC-V, HTML, CSS

Skills: Software Engineering, HPC Optimization, Parallel Computing, Operating Systems, Database, Computer Network

Applications: Git, Vim, Matlab, Latex, Linux Shell, Intel oneAPI Toolkit, Logisim, Spack, Arm Forge, Slurm, Docker,

QEMU, Bochs, ZeroMQ, Kubernetes, Selenium, OpenGL, Oracle, SpringBoot, Maven, JavaFX, AWS EC2, S3

Interests: Software Development, Competitive Programming, System Optimization